KAUSTUV KARAN

+918059645585 ◇ kaustuvkaran@gmail.com ◇ LinkedIn:kaustuvkaran ◇ Github ◇ Portfolio

WORK EXPERIENCE

American Express — Software Development Engineer 2 Jan 2022 - Present

- Boosted turnaround-time of two high-volume APIs by **60% and 83%** by eliminating redundant DB calls and streamlining workflow, resulting in shortening the journey of a user by **10s**.
- Improved performance of 3 React-based services by **90% (15s → 1.57s)** through lazy loading and code refactoring
- Built, maintained and tested a full-stack application using React and Spring Boot as the **founding engineer**.
- Led feature development, conducted code reviews and launched the application with a pilot group of **8000 users**.
- Boosted turnaround time of a different application by **85%** via E2E flow optimization and streamlining API calls.
- Took initiative to implement **Factory Design Pattern** in a service to enhance scalability, optimized exception handling, and **added logging, metrics, and anomaly detection** for performance and throughput.
- **Refactored 100+ Java classes** by replacing boilerplate code with Lombok annotations, reducing codebase size by **3,400 lines** and improving readability, maintainability, and standardization across 3 services.
- Used **Github Co-pilot (AI coding tool)** to mitigate **70+ vulnerabilities** in the codebase, ensuring compliance with secure coding standards
- Leveraged a customised a **GPT-based workflow** to analyze front-end performance traces and bundle metrics, identifying optimization opportunities in React applications to **improve load times and user experience.**
- **Prototyped AI-driven test case generation** for REST APIs using service contracts and controller logic, improving test coverage and reducing manual test authoring effort for backend services.
- **Saved 100+ hours/year** by automating dependency upgrades and deployments across 25 repositories.
- **Technologies Used: React.js, Redux, Jest, SpringBoot, JUnit5, Couchbase, Elasticsearch**

Samsung PRISM Program — Researcher May 2021 - December 2021

- Designed and implemented an **end-to-end facial image reconstruction pipeline** using **GANs and OpenCV** focused on restoring masked facial regions from input images.
- Curated and preprocessed a large-scale facial image dataset (CelebA) by cleaning, augmenting with synthetic face masks, and binarizing masked regions, then tuned a pre-trained GAN model on original and augmented datasets to generate realistic reconstructed faces.

PROJECTS

Sewage Treatment Plant Monitoring System
Ministry of Water Resources, Govt. of India Github

- Map based interface built using Leaflet API for displaying and performing CRUD operations on STP data
- Separation of privilege via role based authentication using passport.js in Super Admin, Admin and Employee roles
- **Technologies Used :** React.js, Leaflet API, Passport.js, Redux, Sheets API, Express.js, Node.js and MongoDB

CodeRED Github

- Map based website to query locations of toilets and stores selling menstrual supplies, for emergencies.
- Google Maps API used to locate nearest store or restrooms during emergency using browser's geolocation.
- **Technologies Used :** React.js, Redux, Node.js, MongoDB, Google Maps API and Auth0 authentication API

EDUCATION

Bachelor of Engineering in Computer Engineering August 2018 - July 2022
Thapar Institute of Engineering and Technology, Patiala, Punjab
CGPA: 8.9/10.00 —— Received Merit Based Scholarship

ACHIEVEMENTS

American Express - Secure Coding Championship - Ranked 8th / 600 globally October 2023

American Express - GCST Hackathon - Ranked 1st / 800 globally August 2023

Smart India Hackathon - Evaluator - 2025 / Winner(National) - 2019 / Intra College - 2020